FYUGP B.Sc. MATHEMATICS HONOURS

ABSTRACT ALGEBRA I (MAT5CJ302)

Semester V · Core in Major

MODULE 2: SUBGROUPS, CYCLIC GROUPS, HOMOMORPHISMS AND PERMUTATIONS

Past questions

Section A: 1 to 3 marks

Subgroups

1 Define subgroup of a group. **(1 mark)** *November 2018*

2 Write the subgroups of S_3. **(2 marks)** *November 2022*

Cyclic subgroups

3 Find all cyclic subgroups of the group $\mathbb{Z}_6$. **(2 marks)** *November 2022*

4 In $GL_2(\mathbb{R})$, find the order of $\begin{bmatrix} 1 & -1 \\ 1 & 0 \end{bmatrix}$. **(2 marks)** *November 2024*

5 Let G be a group and $a \in G$. Then show that $\langle a \rangle$ is a sub-group of G. **(2 marks)** *November 2025*

Cyclic groups

6 Define a cyclic group and give an example. **(1 mark)** *November 2018*

7 State True or False. “Every abelian group is cyclic”. **(1 mark)** *November 2021*

8 Give an example of non-cyclic group with four elements. **(1 mark)** *November 2021*

9 Write the number of generators of the group $\mathbb{Z}_5$ under addition modulo 5. **(1 mark)** *November 2021*

10 The total number of subgroups of $\mathbb{Z}_{12}$ is _____. **(1 mark)** *November 2021*

11 Is S_3 cyclic ? Justify your answer. **(2 marks)** *November 2023*

12 Give the subgroup diagram of $\mathbb{Z}_{24}$. **(2 marks)** *November 2023*

13 Give the sub-group diagram of $\mathbb{Z}_{60}$. **(2 marks)** *November 2025*

Homomorphisms and kernels

14 What is the necessary condition for a homomorphism ϕ from a group G to G' to be injective. **(1 mark)** *November 2018*

15 Let $\Phi : G_1 \to G_2$ be a group homomorphism. Show that $\Phi(e) = e'$ where e and e' are identity elements of G_1 and G_2 respectively. **(3 marks)** *November 2021*

Permutation groups and Cayley's theorem

16 What are the orbits of the identity permutation σ of a set A ? **(1 mark)** *November 2021*

Section B: 4 to 7 marks

Subgroups

17 Give a necessary and sufficient condition for union of two subgroups of a group to be a subgroup. **(7 marks)** *November 2018*

18 Let G be a group and let H be a subset of G. Then show that H is a subgroup of G if and only if H is non-empty and $ab^{-1} \in H$ for all $a, b \in H$. **(5 marks)** *November 2022*

Cyclic subgroups

19 Consider the group $\mathbb{Z}_{12}$, under the operation addition modulo 12. Find the order of the cyclic subgroup generated by $3 \in \mathbb{Z}_{12}$. **(4 marks)** *November 2021*

20 What is the order of the cycle $(1, 4, 5, 7)$ in S_8 ? **(4 marks)** *November 2021*

21 Find all cyclic subgroups of $\mathbb{Z}_8$. **(5 marks)** *November 2022*

22 Show that the cyclic group $\langle i \rangle$ is isomorphic to $\mathbb{Z}_4$. **(5 marks)** *November 2022*

23 Find the cyclic subgroup generated by $\begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix}$ in $GL_2(\mathbb{Z}_3)$. **(5 marks)** *November 2023*

Cyclic groups

24 Can the identity element be a generator of a cyclic group ? **(4 marks)** *November 2021*

25 Prove that every cyclic group is abelian. **(4 marks)** *November 2021*

26 Find the number of generators of the cyclic group of order 8. **(4 marks)** *November 2021*

27 Show that the subgroups of $\mathbb{Z}$ under addition are precisely the groups $n\mathbb{Z}$ under addition for $n \in \mathbb{Z}$. **(7 marks)** *November 2021*

28 Let G be a finite cyclic group with n elements. Show that $G \cong \mathbb{Z}_n$. **(5 marks)** *November 2022*

29 Give the subgroup diagram of $\mathbb{Z}_{12}$. **(5 marks)** *November 2023*

30 Prove that every subgroup of a cyclic group is cyclic. **(5 marks)** *November 2024*

31 Let G be an infinite cyclic group. Show that $G \cong \mathbb{Z}$. **(5 marks)** *November 2025*

Homomorphisms and kernels

32 Let G and G' be groups and let $\Phi : G \to G'$ be one to one function such that $\Phi(xy) = \Phi(x)\Phi(y)$ for all $x, y \in G$. Then prove that $\Phi[G]$ is a subgroup of G' and Φ provides an isomorphism of G with $\Phi[G']$. *[as printed]* **(7 marks)** *November 2018*

33 Show that a homomorphism ϕ of a group G is a one-to-one function if and only if $\mathbf{Ker}\,\phi = \{e\}$. **(7 marks)** *November 2021*

34 Let G be an abelian group, and let n be any positive integer. Show that the function $\phi : G_1 \to G_2$ defined by $\phi(x) = x^n$ is a homomorphism. *[as printed]* **(5 marks)** *November 2022*

35 Let $\phi : G_1 \to G_2$ be a group homomorphism, with $K = \mathbf{ker}\,\phi$. Prove that K is a subgroup of G_1. **(5 marks)** *November 2023*

36 Let $\phi : G_1 \to G_2$ be a group homomorphism. If H_1 is a sub-group of G_1, then show that $\phi(H_1)$ is a sub-group of G_2. **(5 marks)** *November 2025*

Permutation groups and Cayley's theorem

37 Show that any permutation of a finite set of at least two elements is a product of transpositions. **(7 marks)** *November 2021*

Section C: 10 marks and above

Cyclic groups

38 Show that subgroup of a cyclic group is cyclic. **(13 marks)** *November 2021*

39 Let G be a cyclic group. Show that : (a) If G is infinite then $G \cong \mathbb{Z}$. (b) If $|G| = n$, then $G \cong \mathbb{Z}_n$. **(10 marks)** *November 2021*

Permutation groups and Cayley's theorem

40 State and prove Cayley’s theorem in detail. **(13 marks)** *November 2018*

41 Show that every group is isomorphic to a group of permutations. **(10 marks)** *November 2025*